signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
threshold = 1e-6 * max(magnitude_X_f);

for k = 1:N
    if magnitude_X_f(k) < threshold then
        phase_X_f(k) = 0;
    end
end

// 7. Plot original rectangular waveform - BLUE
scf(0);
clf();

subplot(3,1,1);
plot(t, x_t, "b");
xgrid();
xtitle("Rectangular Waveform", "t (Time)", "Amplitude");

// 8. Plot magnitude spectrum - RED
subplot(3,1,2);
plot(f, magnitude_X_f, "r");
xgrid();
xtitle("Magnitude of the Fourier Transform", ...
       "f (Frequency)", "|X(f)|");

// 9. Plot phase spectrum - GREEN
subplot(3,1,3);
plot(f, phase_X_f, "g");
xgrid();
xtitle("Phase of the Fourier Transform", ...
       "f (Frequency)", "Phase of X(f)");

```

